6.5 THE BOOST CONVERTER

The boost converter is shown in Fig. 6-8. This is another switching converter that operates by periodically opening and closing an electronic switch. It is called a boost converter because the output voltage is larger than the input.

Voltage and Current Relationships

The analysis assumes the following:

1. Steady-state conditions exist.
2. The switching period is T, and the switch is closed for time DT and open for $(1-D)T$.
3. The inductor current is continuous (always positive).
4. The capacitor is very large, and the output voltage is held constant at voltage V_o.
5. The components are ideal.

The analysis proceeds by examining the inductor voltage and current for the switch closed and again for the switch open.

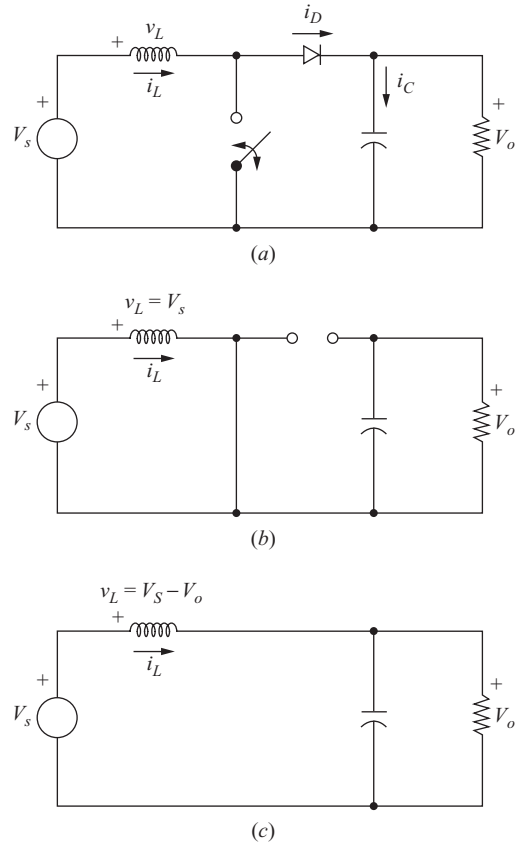


Figure 6-8 The boost converter. (*a*) Circuit; (*b*) Equivalent circuit for the switch closed; (*c*) Equivalent circuit for the switch open.

Analysis for the Switch Closed When the switch is closed, the diode is reverse-biased. Kirchhoff's voltage law around the path containing the source, inductor, and closed switch is

$$v_L = V_s = L\frac{di_L}{dt} \qquad \text{or} \qquad \frac{di_L}{dt} = \frac{V_s}{L} \tag{6-24}$$

The rate of change of current is a constant, so the current increases linearly while the switch is closed, as shown in Fig. 6-9*b*. The change in inductor current is computed from

$$\frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{DT} = \frac{V_s}{L}$$

Solving for Δi_L for the switch closed,

$$(\Delta i_L)_{\text{closed}} = \frac{V_s DT}{L} \tag{6-25}$$

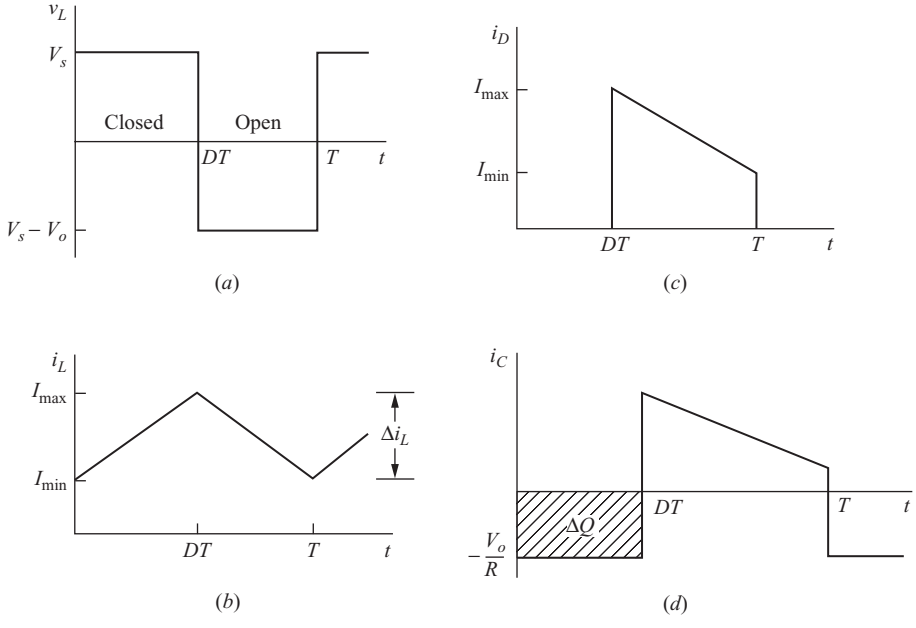


Figure 6-9 Boost converter waveforms. (*a*) Inductor voltage; (*b*) Inductor current; (*c*) Diode current; (*d*) Capacitor current.

Analysis for the Switch Open When the switch is opened, the inductor current cannot change instantaneously, so the diode becomes forward-biased to provide a path for inductor current. Assuming that the output voltage V_o is a constant, the voltage across the inductor is

$$v_L = V_s - V_o = L\frac{di_L}{dt}$$

$$\frac{di_L}{dt} = \frac{V_s - V_o}{L}$$

The rate of change of inductor current is a constant, so the current must change linearly while the switch is open. The change in inductor current while the switch is open is

$$\frac{\Delta i_L}{\Delta t} = \frac{\Delta i_L}{(1-D)T} = \frac{V_s - V_o}{L}$$

Solving for Δi_L,

$$(\Delta i_L)_{\text{open}} = \frac{(V_s - V_o)(1-D)T}{L} \tag{6-26}$$

For steady-state operation, the net change in inductor current must be zero. Using Eqs. (6-25) and (6-26),

$$(\Delta i_L)_{\text{closed}} + (\Delta i_L)_{\text{open}} = 0$$

$$\frac{V_s DT}{L} + \frac{(V_s - V_o)(1 - D)T}{L} = 0$$

Solving for V_o,

$$V_s(D + 1 - D) - V_o(1 - D) = 0$$

$$\boxed{V_o = \frac{V_s}{1 - D}} \tag{6-27}$$

Also, the average inductor voltage must be zero for periodic operation. Expressing the average inductor voltage over one switching period,

$$V_L = V_s D + (V_s - V_o)(1 - D) = 0$$

Solving for V_o yields the same result as in Eq. (6-27).

Equation (6-27) shows that if the switch is always open and D is zero, the output voltage is the same as the input. As the duty ratio is increased, the denominator of Eq. (6-27) becomes smaller, resulting in a larger output voltage. *The boost converter produces an output voltage that is greater than or equal to the input voltage.* However, the output voltage cannot be less than the input, as was the case with the buck converter.

As the duty ratio of the switch approaches 1, the output voltage goes to infinity according to Eq. (6-27). However, Eq. (6-27) is based on ideal components. Real components that have losses will prevent such an occurrence, as shown later in this section. Figure 6-9 shows the voltage and current waveforms for the boost converter.

The average current in the inductor is determined by recognizing that the average power supplied by the source must be the same as the average power absorbed by the load resistor. Output power is

$$P_o = \frac{V_o^2}{R} = V_o I_o$$

and input power is $V_s I_s = V_s I_L$. Equating input and output powers and using Eq. (6-27),

$$V_s I_L = \frac{V_o^2}{R} = \frac{[V_s/(1 - D)]^2}{R} = \frac{V_s^2}{(1 - D)^2} R$$

By solving for average inductor current and making various substitutions, I_L can be expressed as

$$\boxed{I_L = \frac{V_s}{(1-D)^2 R} = \frac{V_o^2}{V_s R} = \frac{V_o I_o}{V_s}} \tag{6-28}$$

Maximum and minimum inductor currents are determined by using the average value and the change in current from Eq. (6-25).

$$I_{\max} = I_L + \frac{\Delta i_L}{2} = \frac{V_s}{(1-D)^2 R} + \frac{V_s DT}{2L} \tag{6-29}$$

$$I_{\min} = I_L - \frac{\Delta i_L}{2} = \frac{V_s}{(1-D)^2 R} - \frac{V_s DT}{2L} \tag{6-30}$$

Equation (6-27) was developed with the assumption that the inductor current is continuous, meaning that it is always positive. A condition necessary for continuous inductor current is for $I_{\min}$ to be positive. Therefore, the boundary between continuous and discontinuous inductor current is determined from

$$I_{\min} = 0 = \frac{V_s}{(1-D)^2 R} - \frac{V_s DT}{2L}$$

or

$$\frac{V_s}{(1-D)^2 R} = \frac{V_s DT}{2L} = \frac{V_s D}{2Lf}$$

The minimum combination of inductance and switching frequency for continuous current in the boost converter is therefore

$$(Lf)_{\min} = \frac{D(1-D)^2 R}{2} \tag{6-31}$$

or

$$\boxed{L_{\min} = \frac{D(1-D)^2 R}{2f}} \tag{6-32}$$

A boost converter designed for continuous-current operation will have an inductor value greater than $L_{\min}$.

From a design perspective, it is useful to express L in terms of a desired Δi_L,

$$L = \frac{V_s DT}{\Delta i_L} = \frac{V_s D}{\Delta i_L f} \tag{6-33}$$

Output Voltage Ripple

The preceding equations were developed on the assumption that the output voltage was a constant, implying an infinite capacitance. In practice, a finite capacitance will result in some fluctuation in output voltage, or ripple.

The peak-to-peak output voltage ripple can be calculated from the capacitor current waveform, shown in Fig. 6-9*d*. The change in capacitor charge can be calculated from

$$|\Delta Q| = \left(\frac{V_o}{R}\right) DT = C\Delta V_o$$

An expression for ripple voltage is then

$$\Delta V_o = \frac{V_o DT}{RC} = \frac{V_o D}{RCf}$$

or

$$\boxed{\frac{\Delta V_o}{V_o} = \frac{D}{RCf}} \tag{6-34}$$

where f is the switching frequency. Alternatively, expressing capacitance in terms of output voltage ripple yields

$$C = \frac{D}{R(\Delta V_o / V_o) f} \tag{6-35}$$

As with the buck converter, equivalent series resistance of the capacitor can contribute significantly to the output voltage ripple. The peak-to-peak variation in capacitor current (Fig. 6-9) is the same as the maximum current in the inductor. The voltage ripple due to the ESR is

$$\Delta V_{o,\text{ESR}} = \Delta i_C r_C = I_{L,\max} r_C \tag{6-36}$$

EXAMPLE 6-4

Boost Converter Design 1

Design a boost converter that will have an output of 30 V from a 12-V source. Design for continuous inductor current and an output ripple voltage of less than one percent. The load is a resistance of 50 Ω. Assume ideal components for this design.

■ Solution

First, determine the duty ratio from Eq. (6-27),

$$D = 1 - \frac{V_s}{V_o} = 1 - \frac{12}{30} = 0.6$$

If the switching frequency is selected at 25 kHz to be above the audio range, then the minimum inductance for continuous current is determined from Eq. (6-32).

$$L_{\min} = \frac{D(1-D)^2(R)}{2f} = \frac{0.6(1-0.6)^2(50)}{2(25{,}000)} = 96\ \mu\text{H}$$

To provide a margin to ensure continuous current, let $L = 120\ \mu$H. Note that L and f are selected somewhat arbitrarily and that other combinations will also give continuous current.

Using Eqs. (6-28) and (6-25),

$$I_L = \frac{V_s}{(1-D)^2(R)} = \frac{12}{(1-0.6)^2(50)} = 1.5\ \text{A}$$

$$\frac{\Delta i_L}{2} = \frac{V_s DT}{2L} = \frac{(12)(0.6)}{(2)(120)(10)^{-6}(25{,}000)} = 1.2\,\text{A}$$

$$I_{\max} = 1.5 + 1.2 = 2.7\,\text{A}$$

$$I_{\min} = 1.5 - 1.2 = 0.3\,\text{A}$$

The minimum capacitance required to limit the output ripple voltage to 1 percent is determined from Eq. (6-35).

$$C \geq \frac{D}{R(\Delta V_o / V_o) f} = \frac{0.6}{(50)(0.01)(25{,}000)} = 48\ \mu\text{F}$$